

Data Exploration and Preparation

Diploma in DAB – L7

CCT College Dublin

Introduction to Data Exploration

Week 1

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Outline

- Introduction to Data
- Variables, observations and attributes in Data Analytics
- Data types
- Primary and Secondary data
- Steps involved in data acquisition and management
- Version control system
- Digital tools



Introduction to Data

- **Data** consists of recorded facts, measurements, observations or events that can be stored, processed and analysed.
- **Examples:** Customer age, Product price, Number of purchases, Transaction dates, Images, Social-media posts and Sensor readings

Data vs Information

- **Data:** Raw observations or recorded values.
- **Information:** Data that has been organised, processed or interpreted to provide meaning.
- **Data:** €120, €95, €180, €210, €140 and
- **Information:** Average customer transaction = €149

Understanding a Dataset

Customer ID	Age	Region	Purchases	Spend
C001	24	Dublin	5	€320
C002	31	Cork	3	€190
C003	42	Galway	8	€640

- **Observation:** An observation represents one individual entity or instance being studied.
- For example: One row = one customer observation
- **Variable:** A variable represents something that can take different values.
- For example: Age, Region, Number of purchases, Spend
- **Attribute:** An attribute is a characteristic or property describing an entity.
- For example, Age and Region are attributes of a customer.
- In analytics, the terms variable, feature and attribute may sometimes be used in closely related ways.

Types of Data

- Understanding the type of data is important because it determines which analytical techniques can be applied.
- **Categorical Data:** Represents groups or labels. For example, Department, Country, Customer type and Product category
- **Nominal:** Categories without a natural order. For example: Dublin, Cork, Galway
- **Ordinal:** Categories with a meaningful order. For example: Low → Medium → High
- **Numerical Data:** Represents measurable quantities.
- **Discrete Data:** Usually countable values. For example: Number of purchases = 6
- **Continuous Data:** Can take values within a range. For example: Revenue = €1,245.75
- Is Customer Satisfaction: Poor / Average / Good

Data Management

- Data management concerns how data is
- **Collected → Stored → Organised → Maintained → Used → Shared → Archived**
- Good data management makes data easier to **Locate, Understand, Analyse, Reproduce, Protect and Share appropriately.**
- Important Data Management Practices are
  - **File organization:** Consistent folder structures and Meaningful file names
  - **Documentation:** Data dictionaries, README files and Metadata
  - **Data quality:** Accuracy, Completeness, Consistency and Validity
  - **Data provenance:** Where did the data originate? When was it collected? How was it transformed?
  - **Principle:** Good analysis begins with well-managed and well-understood data.

Primary and Secondary Data

- **Primary Data:** Data collected directly for a particular research or business purpose.
- **Examples:** Surveys, Interviews, Experiments, Customer questionnaires, Sensor measurements and Observation
- **Advantages:** Specific to the research objective, Greater control over collection and Can provide current information
- **Challenges:** Time consuming, Can be expensive and Requires careful design

- **Secondary Data:** Data that has already been collected by another organization or for another purpose.
- **Examples:** Government open-data portals, Central Statistics Office datasets, Company databases, Published reports, Public datasets and Research repositories
- **Advantages:** Usually quicker to obtain, Often inexpensive and Can contain large datasets
- **Challenges:** Quality may be unknown, Data may be outdated and Definitions may differ from your requirements

What, Where and How of Data Acquisition

- Before acquiring data, an analyst should ask three questions.

WHAT data do we need?

- Start with the business question.
- Example: What factors influence customer churn?
- Possible required variables: Customer age, Contract type, Monthly charges, Customer service interactions, Length of relationship and Churn status

WHERE can we obtain the data?

- Possible sources include: Internal company databases, CSV/Excel files, APIs, Government open-data portals, Surveys, Public datasets, Sensors and Web-based sources

HOW will we acquire it?

- Possible methods: File download, Database query, API request, Survey, Data collection system and Approved web-data collection
- Before collecting data, also consider: **Privacy + Ethics + Licensing + Bias + Quality + Security**

Version Control

Git and GitHub

- **Why Version Control?**
- During an analytics project, datasets, Python files and notebooks change repeatedly.
- Without version control, files can become: analysis_final.py, analysis_final2.py, analysis_final_revised.py and analysis_really_final.py
- Git provides a systematic approach.
- **Important Git Concepts**
- **Repository:** Folder containing the project and its version history.
- **Commit:** A recorded snapshot of project changes.
- **Push:** Send local commits to GitHub.
- **Pull:** Retrieve changes from a remote repository.
- **Branch:** A separate development path used to work on changes safely.
- **Why use GitHub in Data Analytics?**
  - Tracks project development
  - Provides evidence of work
  - Supports collaboration
  - Protects against accidental loss
  - Improves reproducibility
  - Builds professional practice
  - Assignment Practice

Resources/ References

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